Fiscal Year:	FY 2022	**Task Last Updated:**	FY 04/04/2022
PI Name:	Rivas, Eric Ph.D.		
Project Title:	Temporal Changes in Astronauts' Muscle and Cardiorespiratory Physiology Pre, During, and Post Spaceflight		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	eric.rivas@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	
Organization Name:	KBR/NASA Johnson Space Center		
PI Address 1:	Human Physiology, Performance, Protection, and Operations Laboratory		
PI Address 2:	2400 NASA Parkway, JSC B21		
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	36
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2017-2018 HERO 80JSC017N0001-BPBA Topics in Biological, Physiological, and Behavioral Adaptations to Spaceflight. Appendix C
Start Date:	06/07/2021	**End Date:**	09/30/2033
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
Contact Email:	becky.brocato@nasa.gov		
Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:	March 2023 Update: Dr. Rivas is no longer with the project. The new Principal Investigator (PI) as of March 2023 is Nicole Strock, Ph.D. Dr. Eric Rivas had joined the project as PI in June 2021; previous PI was Dr. Meghan Downs. Collaborators are Scott Smith, Ph.D. and Jason Norcross, M.S.		
COI Name (Institution):	Barstow, Thomas Ph.D. (Kansas State University) Ade, Carl Ph.D. (Kansas State University) Martin, David M.S. (KBR/NASA Johnson Space Center) Abercromby, Andrew Ph.D. (NASA Johnson Space Center) Young, Millennia Ph.D. (NASA Johnson Space Center)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Continuation of "Temporal Changes in Astronauts' Muscle and Cardiorespiratory Physiology Pre, During, and Post Spaceflight" with new Principal Investigator, Dr. Eric Rivas. Quantification of astronauts' changes in cardiorespiratory fitness and muscle performance and size in parallel with monitoring of pre and in-flight lifestyle habits (i.e., exercise training, nutritional intake, and sleep patterns) is needed to develop countermeasures and technologies for monitoring and mitigating crew health and performance risks during exploration class missions. The research proposed herein will temporally profile changes in astronauts' cardiorespiratory fitness and muscle mass, strength, and endurance over the course of spaceflight missions ranging from 2 months, 6 months, and up to 1 year in duration. Additionally, a statistical based extrapolation will provide predictions for changes associated with exploration missions 2-3 years in duration. To accomplish these objectives, lower and upper body muscle strength, power, and endurance will be measured using a well validated test battery consisting of leg extension, leg press, isokinetic, bench press tests, and isometric mid-thigh pull test. Muscle size will be assessed pre, in, and post-flight using well validated magnetic resonance imaging (MRI) and ultrasound imaging techniques. Cardiorespiratory fitness and related parameters will be tested pre, in, and post-flight using traditional VO2peak test and critical power test protocols paired with non-invasive assessments of oxygen consumption, cerebral and muscle oxygenation and perfusion. Ambulatory and in-flight exercise, nutrition, and sleep will be monitored using a variety of commercial technologies and in-flight assessment tools. This proposal specifically addresses the temporal effects of spaceflight on changes in cardiorespiratory fitness and muscle function, both critical parameters in maintaining the ability to perform mission critical tasks and enabling safe human space exploration beyond low Earth orbit. Integration of data collected during the pre and in-flight periods will lead to a better understanding of how to optimize exercise and non-exercise countermeasures to maintain crew health, safety, and performance during the exploration era.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	It is well established exercise has many health benefits. This research will improve our understanding about the variability fitness improvements and how they change over time. Understanding this will improve exercise prescriptions for athletic and clinical populations on Earth.
Task Progress:	NOTE: P.I. changed to Nicole Strock, Ph.D effective March 2023. For subsequent reporting, please see "Temporal Changes in Astronauts' Muscle and Cardiorespiratory Physiology Pre, During, and Post Spaceflight (PI: Strock)". (Ed. 3/29/23) Completed task include institutional review board protocol and consent form modification approvals, test readiness review for B21 and B26, and informed consent briefing for USCV-4. (Ed. Note: USCV-4 refers to the fourth operational flight of crewed spacecraft launched on a SpaceX Falcon 9 rocket on 4/27/2022 (also known as USCV-4 or Crew-4).
Bibliography Type:	Description: (Last Updated:)